

CHAPTER

30



Shitapitta, Udarda and Kotha

Learning Objectives

- ⇒ To learn the concept of Shitapitta
- ⇒ To learn the Nidana Panchaka of Shitapitta mentioned in Ayurveda literatures
- ⇒ To frame out the probable Samprapti Ghataka based on progression of the disease
- ⇒ To learn about frostbite, chilblain and immersion (trench) foot

Introduction

Shitapitta, Udarda and Kotha are spectra of diseases where we can observe skin eruptions influenced by environmental factors, lifestyle and sensitivity. Though, these sets of diseases are involving skin abnormalities, these are not included in *Kushtha* disease spectrum because of specific etiological factors and specific clinical features with atypical progression of the diseases even though the *Tvak* (skin) is afflicted and *Rasa* and *Rakta Dhatu* is involved during the progression of the disease.

Shitapitta

शीतमारुतसंस्पर्शात्प्रदुष्टौ कफमारुतौ।

पित्तेन सह संभूय बहिरन्तर्विसर्पतः॥१॥

Ma.Ni. - 50/1

- ⊗ *Shitapitta* is a spectrum of disease where in the person develops eruptions in the **skin**¹.
- ⊗ This is caused by the exposure of the **skin to the cold environment**².
- ⊗ This is having the abnormal vitiation of *Pitta Dosh*

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- 1 Peripheral cold injuries include both freezing and non freezing injuries to the tissues. Tissue freezes quickly when in contact with thermal conductors such as metal or volatile solutions. Other predisposing factors include constrictive clothing or boots, immobility or vasoconstrictive medications.
Frostbite occurs when the tissue temperature falls below 0° C. Ice crystal formation distorts and destroys cellular architecture. Clinically, frostbite is classified as superficial or deep. Superficial frostbite does not entail tissue loss. Classically, frostbite is classified like a burn retrospectively. Depending upon the degree of freezing injuries, four different grading of freeze injuries are classified. 1st degree frostbite causes only erythema and oedema. The appearance of superficial vesiculation surrounded by oedema and erythema is considered as 2nd degree. In 3rd degree frostbite shows haemorrhagic vesicles. 4th degree frostbite injuries damage subcuticular, muscular and osseous tissues.
The two most common non freezing injuries include chilblain (pernio) and immersion (trench) foot. Chilblain results from neuronal and endothelial damage induced by repetitive exposure to dry cold. Young females particularly with a history of Raynaud's phenomenon are at higher risk for chilblain. Persistent vasospasticity and vasculitis can cause erythema, mild oedema and pruritus. Eventually, plaques, blue nodules and ulcerations develop.
Immersion (trench) foot develops from repetitive exposure to wet cold above the freezing point. The feet initially appear cyanotic, cold and oedematous, subsequently leads to the formation of bullae and later liquefaction necrosis.
- 2 As the person is exposed to Shita Maruta (cold wind), that will lead to the increase of Shita Guna externally as well as internally. Shita Guna is the common quality observed for Vata and Kapha Dosh. That is the reason, when a person is exposed to Shita quality, there will be increase of Vata and Kapha Dosh is likely.
When an individual is exposed to extreme cold environment, surface area of the human body can freeze. This is called frostbite. This occurs especially in the ear lobes and in the digits of hands and feet. If the freezing is sufficient to cause extensive formation of ice crystals in the cells, permanent damage usually occurs, such as permanent circulatory impairment as well as local tissue damage. Gangrene will be sequel of this pathological event. In this situation, there will be peripheral vaso constriction as well as skeletal muscle contraction. This will be associated with piloerection means hairs standing on the end. This occurs due to the stimulation of sympathetic nervous system and there will be increased thermogenesis due to sympathetic excitation heat production and thyroxine secretion.

along with the association of *Kapha and Vata Dosha*³.
 * The clinical features can be observed both internally and externally⁴.

Purvarupa of Shitapitta

पिपासारुचिहल्लासदेहसादाङ्गौरवम्।

रक्तलोचनता तेषां पूर्वरूपस्य लक्षणम्॥12॥

Ma.Ni. – 50/2

- * **Pipasa**⁵ – thirst
- * **Aruchi**⁶ – tastelessness
- * **Hrillasa** – nausea
- * **Dehasada** – sluggish movement of the body parts
- * **Anga gourava**⁷ – heaviness of the body parts
- * **Rakta lochanata**⁸ – reddish discolouration of the eyes

Lakshana of Shitapitta and Udarda⁹

वरीष्टसंस्थानः शोथः संजायते बहिः।

सकण्डूस्तोदबहुलइच्छर्दिज्वरविदाहवान्॥13॥

उदरमिति तं विद्याच्छीतपित्तमथापरे।

वाताधिकं शीतपित्तमुदरस्तु कफाधिकः॥14॥

Ma.Ni. – 50/3,4

- * **Varati Damshta Samsthana Shotha**¹⁰ – patchy elevation observed in the skin
- * **Kandu**¹¹ – itching
- * **Toda Bahula**¹² – pricking type of pain
- * **Chardi**¹³ – vomiting
- * **Jvara**¹⁴ – raised body temperature
- * **Vidaha**¹⁵ – burning sensation in the area of skin eruption

Above said *Lakshana* observed along with vitiated *Vata*, is termed as *Shitapitta* as per one author and same *Lakshana* seen along with vitiated *Kapha Dosha* is termed as *Udarda*.

Udarda

सोत्सङ्गैश्च सरागैश्च कण्डूमद्भिश्च मण्डलैः।

शैशिरः कफजो व्याधिरुदरं इति कीर्तितः॥15॥

Ma.Ni. – 50/5

Once the vascular endothelium is damaged, stasis of the blood occurs rapidly to microvascular thrombosis. The microvasculature begins to collapse, arteriovenous shunting increases tissue pressures and oedema forms. Finally, thrombosis, ischemia and superficial necrosis appear.

3 वाताहते नास्ति रुजा न पाकः पित्ताते नास्ति कफाच्च पूयः॥ तस्मात् समस्तान् परिपाककाले पचन्ति शोफांस्त्रय एव दोषाः॥18॥

कालान्तरेणाभ्युदितं तु पित्तं कृत्वा वशे वातकफौ प्रसह्य॥ पचत्यतः शोणितमेव पाको मतोऽपरेषां विदुषां द्वितीयः॥19॥

Su.Su. – 17/8,9

The above said verse represents the Dosha vitiation in the body and its impact on the development of pathological events. When the person is having any kind of pain, it indicates the involvement of Vata Dosha. When the person is showing the stages of suppuration, it indicates the vitiation of Pitta Dosha. When the person is developing the pus formation during the course of the disease, it suggests the Kapha Dosha involvement during the progression of the disease. When the person is having the progression of pathological events in Vranashotha (inflammation), we can predict the involvement of Rakta during the pathogenesis.

When the person is exposed to Shita environment continuously for longer time, there will be formation of the gangrene in the lower limb. When we observe the gangrenous changes, it implies the rapidity of development of symptoms (Pitta Dosha involvement) and pus formation (Kapha Dosha vitiation).

4 When the person is exposed to ice cold water for a period of 20 to 30 minutes, the person dies because of cardiac arrest or atrial fibrillation. The person can also have the development of freezing of the ear lobes and digits of the hands and feet.

5 This can be attributable to sympathetic nervous system activation. This will cause peripheral vaso constriction to prevent the heat loss from the body. As there is sympathetic nervous system activation, there will be inhibition of glands of gastro intestinal tract, thus the person may have manifestation of thirst due to inhibition of salivary secretion.

6 This can be attributable to again sympathetic nervous system stimulation leading to the poor secretion from the glands of alimentary canal. As the person will have poor secretion from the mouth and stomach, he will likely to have tastelessness.

7 Due to the peripheral vaso constriction, there will be stasis of the blood in the intravascular compartment, interstitial compartment. This mechanism leads to the feeling of the heaviness of the body parts, thus the person may have sluggish movement of body parts.

8 The term Rakta Lochanata refers to the reddish discolouration of the eyes. Eye is considered to be one of the important organ to be exposed to the cold environment apart from skin, ears etc. Thus, continuous exposure to cold environment leads to the damage to the peripheral blood vessels supplying the ocular region leading to the reddish discolouration of the eyes.

9 Based on the similarity between the pathological events happening between Shitapitta and Udarda, there will be bleeding tendency below the skin. Depending upon the size of the bleeding spots observed across the skin determines the diagnosis of Shitapitta and Udarda.

10 When the person is having peripheral vaso constriction, subcutaneous bleeding can be manifested. Superficial crops of small capillary haemorrhages can be referred as petechiae. These are best seen in skin, eyes and mucous membranes. Ecchymosis is the term used in case of confluence of petechiae or haemorrhages from larger blood vessels. Multiple bruises in the form of ecchymosis or haematoma indicates the bleeding diathesis and presence of petechiae indicates the abnormality of platelets and vessel wall. In this context, you have to consider the abnormality of vessel wall.

11 This clinical feature is observed in one of non freezing injury developed by exposure to dry cold. This symptom occurs due to persistent vasospasticity and vasculitis.

12 The presence of pricking type of pain indicates inflammatory reaction happening at the site of exposure to cold wind.

13 Vomiting observed in the patient suffering from the clinical condition can be attributable to the sepsis manifests after the appearance of necrosis.

14 The presence of fever indicates either the inflammation or infection occurring at the site of the cold exposure.

15 Burning sensation is observed in chilblain which manifests due to neuronal damage at the site of non freezing injury.

- ❖ *Udarda*¹⁶ is a spectrum of skin eruption where there is vitiation of *Kapha Dosha*.
- ❖ This is associated with circular, reddish, skin eruptions or elevations along with profuse itching.
- ❖ This is more common in *Shishira Rtu* (winter season).

Description of Utkotha and Kotha

असम्यग्बमनोदीर्णपित्तश्लेष्मान्निग्रहेः।

मण्डलानि सकण्डूनि रागवन्ति बहूनि च।

उत्कोठः सानुबन्धश्च कोठ इत्यभिधीयते॥६॥ Ma.Ni. – 50/6

- ❖ *Utkotha* is a spectrum of skin eruptions which manifests due to improper *Vamana (emesis) therapy*¹⁷.
- ❖ There will be vitiation of *Pitta, Kapha Dosha* and person will be avoiding consumption of food.
- ❖ The person will be presented with multiple, reddish, circular lesions associated with itching. This is also being termed as *Kotha*.
- ❖ The difference between *Kotha* and *Utkotha* is in recurrence. If the person is presented with skin eruptions once, then such type of disease is termed as *Kotha* and if the person is presented with repeated recurrences of skin eruptions is termed as *Utkotha*.



Figure 118: Formation of Kotha or necrosis



Figure 119: Image of chilblain



Figure 120: Image of trench foot

To summarise, this spectrum of diseases is characterized by clinical manifestation developed due to absent or poor peripheral blood supply. This can usually occur in either in upper or lower limbs. This is identified easily as these are visible in the integumentary system. Similar pathological events can occur inside human body which can result in *Shula* or *Gulma* or *Hridroga* depending upon the site of the lesion which can be identified by appropriate laboratory investigations.

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16 *Udarda* can be attributable to the skin manifestations observed after the exposure to cold atmosphere.

17 As per Ayurveda literatures, *Vamana (emesis) therapy* is performed when an individual is suffering from extreme vitiation of *Kapha Dosha* and the person is physically fit to perform *Vamana therapy*. Due to improper preoperative procedure such as improper *Snehana* (oleation therapy) and *Svedana* (fomentation therapy) or improper diet and regimen during *Snehana* or *Svedana* or improper timing of *Vamana therapy* or improper dosage of *Vamana drug*, the vitiated *Kapha Dosha* may not move completely to the exterior. This will lead to the oblique movement of vitiated *Dosha* to the periphery.

As per current understanding, if pathogens are not eliminated from the stomach through emesis, these will enter into the intravascular compartment. From intravascular compartment, the pathogens will move into interstitial compartment and ultimately these pathogens can find the place in intracellular compartment. In this situation, intracellular compartment of the skin.

Key points of the chapter

- Shitapitta, Udarda and Kotha are the spectra of the diseases which occur as a result of exposure to cold environment, improper diet and regimen following emesis therapy respectively. In these conditions, the person will have skin manifestations.
- Shitapitta, Udarda and Kotha will be manifested due to the exposure to Shita quality to the Tvak region. There will be affliction of Rasa, Rakta Dhatu during the course of the manifestation of the disease. There will be affliction of Rasavaha and Raktavaha Srotas during the progression of the disease.
- Frostbite, chilblain and immersion (trench) foot are the clinical conditions which may mimic the features of the Shitapitta, Udarda and Kotha.

Probable questions asked in the examination

Long essay questions

- Explain Shitapitta in detail and write the differences between Shitapitta, Udarda and Kotha

Short essays

- Explain Shitapitta, Udarda and Kotha

MCQs

- Dosha dominance observed in Shita Pitta are
 - Vata and Pitta;
 - Vata and Kapha;
 - Pitta and Kapha;
 - Kapha only

Answer: (b)
- The disease which manifests due to the improper emesis therapy is
 - Shitapitta;
 - Udarda;
 - Utkotha;
 - Udavarta

Answer: (c)
- Vata and Kapha vitiation at a particular time is seen in
 - Shitapitta;
 - Raktapita;
 - Amlapitta;
 - Ushnavata

Answer: (a)
- Udarda is observed in which of the following season?
 - Vasanta Rtu;
 - Sharat Rtu;
 - Shishira Rtu;
 - Varsha Rtu

Answer: (c)
- Frostbite is seen in which part of the body?
 - Abdomen;
 - Back
 - Neck;
 - Digits of upper and lower limbs;

Answer: (d)
- The predominant Dosha responsible for the manifestation of Udarda is
 - Vata;
 - Kapha;
 - Pitta;
 - Vata and Kapha

Answer: (b)
- Itching, annular inflammatory red patches dominated with Kapha are the Lakshana of
 - Shitapitta;
 - Kotha;

- Udarda;
- All of the above

Answer: (c)

- Varati Dashta Samsthana is the term explained in the context of

- Shitapitta;
- Kotha;
- Visarpa;
- Udarda

Answer: (a)

- Frequent remission and exacerbation of symptoms are seen in

- Udarda;
- Kotha;
- Shitapitta;
- None of the above

Answer: (c)

- Udbhava Sthana of Shitapitta is

- Amashaya;
- Pakvashaya;
- Both Amashaya and Pakvashaya;
- No Udbhava Sthana is involved in the manifestation of the disease

Answer: (d)

Chapter resources

- Madhava Nidana – 50th chapter
- Golwalla's medicine for students by Aspi F Golwalla and Sharukh A Golwalla, edited by Milind Y Nadkar, 25th edition, Jaypee Brothers Medical Publishers (P) Ltd, New Delhi, 2017
- The Merck Manual of Diagnosis and Therapy by Robert S Porter and Justin L Kaplan, 19th edition, Merck Sharp & Dohme Corp, New Jersey, 2011
- Chamberlain's symptoms and signs in clinical medicine by Colin Ogilvie and Christopher C Evans, 12th edition, Butterworth-Heinemann Publishers, 1997
- Best & Taylor's physiological basis of medical practice by O P Tandon and Y Tripathi, 13th edition, Wolters Kluwer Health (India) Pvt. Ltd, Gurgaon, 2012
- Textbook of Medical physiology by Arthur C Guyton and John E Hall, 11th edition, Saunders Elsevier publications, Philadelphia, 2006
- clinicaladvisor.com, firstderm.com, healthline.com
